Hands-On Activity: Resolve spreadsheet errors

Activity Overview

In this activity, you'll learn how to identify and fix common spreadsheet errors. You'll work with a pre-populated dataset containing errors, then apply methods to correct them. As a data professional, identifying and debugging errors is crucial for accurate data analysis.

Step-By-Step Instructions

Follow the instructions to complete each step of the activity. Then answer the question at the end of the activity before going to the next course item.

Step 1: Access the spreadsheet

To get started, determine which software you'd like to use to create your chart, such as Google Sheets or Microsoft Excel.

Save the spreadsheet with your preferred file naming convention, and store it in a folder to help you stay organized.

To use the template for this course item, click the link below and select "Use Template."

Link to template: Resolve spreadsheet errors spreadsheet

OR

If you don't have a Google account, download the template directly from the attachment below.

Resolve spreadsheet errors

XLSX File

Step 2: Explore the spreadsheet

1. Open the **Resolve spreadsheet errors** spreadsheet.
2. Take a few minutes to explore the first two sheets, **Descriptions** and **Examples**, to learn more about common errors and to explore some examples.
3. At the end of the activity, feel free to explore the **Solutions** sheet to review an explanation of errors and their solutions directly in the sheet.

The following screenshot captures the **Descriptions** sheet that you will find when you open the **Resolve spreadsheet errors** sheet. It contains common error types, example formulas, and a description of each error.

Resolving spreadsheet errors

File Edit View Insert Format Data Tools Extensions Help Accessibility

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	A	B	C	D
1	**Error Type**	**Example Formula**	**Description**	
2	DIV Error	=A1/B1	Division by zero or empty cell	
3	ERROR (Parsing Error)	=SUM(A1 A2)	Syntax issues, missing comma	
4	N/A Error	=VLOOKUP("Apple", A2:B5, 2, FALSE)	Data not found in lookup	
5	NAME Error	=SUMM(A1:A5)	Typo or incorrect function name	
6	NUM Error	=SQRT(-1)	Data doesn't make sense for calculation	
7	VALUE Error	=SUM(A1:A3)	Incompatible data types	
8	REF Error	=A1+B1	Referenced cells have been deleted	

Resolving spreadsheet errors

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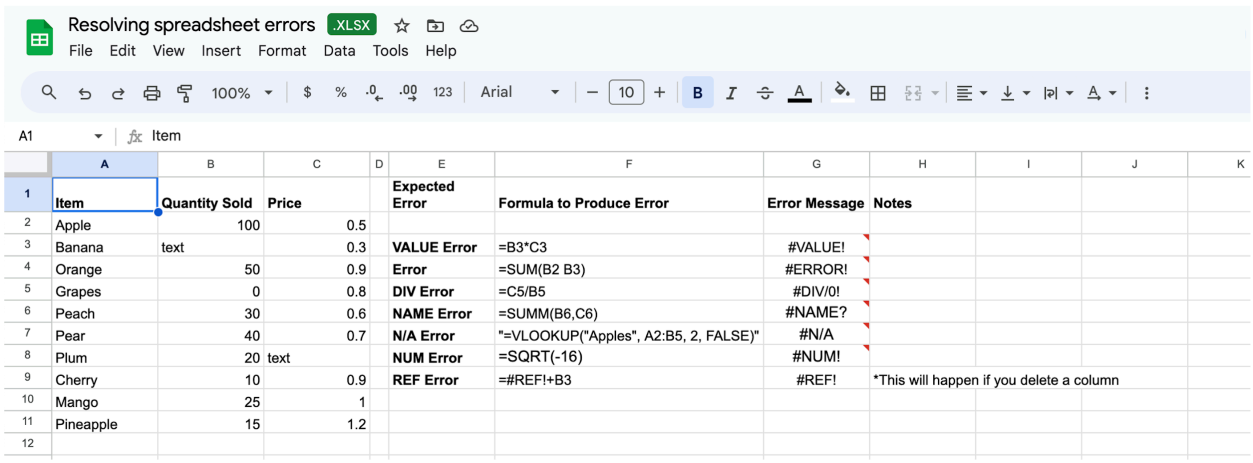
	A	B	C	D	E	F	G	H	I	J
1	**Item**	**Quantity Sold**	**Price**		**Expected Error**	**Formula to Produce Error**	**Error Message**	**Notes**		
2	Apple	100	0.5							
3	Banana	text	0.3		**VALUE Error**	=B3*C3	#VALUE!			
4	Orange	50	0.9		**Error**	=SUM(B2 B3)	#ERROR!			
5	Grapes	0	0.8		**DIV Error**	=C5/B5	#DIV/0!			
6	Peach	30	0.6		**NAME Error**	=SUMM(B6,C6)	#NAME?			
7	Pear	40	0.7		**N/A Error**	"=VLOOKUP("Apples", A2:B5, 2, FALSE)"	#N/A			
8	Plum	20	text		**NUM Error**	=SQRT(-16)	#NUM!			
9	Cherry	10	0.9		**REF Error**	=#REF!+B3	#REF!	*This can happen if you delete a column		
10	Mango	25	1							
11	Pineapple	15	1.2							

Step 3: Identify and resolve common errors

Error types and examples

In the **Resolve spreadsheet errors** spreadsheet, select the **Fix the Examples Here** sheet.

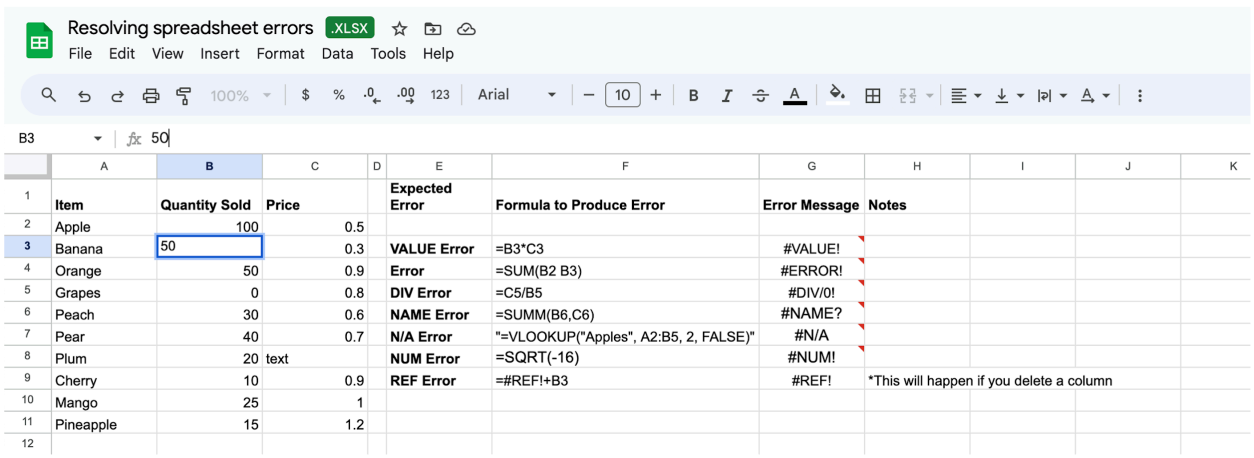


	A	B	C	D	E	F	G	H	I	J	K
1	**Item**	**Quantity Sold**	**Price**		**Expected Error**	**Formula to Produce Error**	**Error Message**	**Notes**			
2	Apple	100	0.5								
3	Banana	text	0.3		**VALUE Error**	=B3*C3	#VALUE!				
4	Orange	50	0.9		**Error**	=SUM(B2 B3)	#ERROR!				
5	Grapes	0	0.8		**DIV Error**	=C5/B5	#DIV/0!				
6	Peach	30	0.6		**NAME Error**	=SUMM(B6,C6)	#NAME?				
7	Pear	40	0.7		**N/A Error**	"=VLOOKUP("Apples", A2:B5, 2, FALSE)"	#N/A				
8	Plum	20	text		**NUM Error**	=SQRT(-16)	#NUM!				
9	Cherry	10	0.9		**REF Error**	=#REF!+B3	#REF!	*This will happen if you delete a column			
10	Mango	25	1								
11	Pineapple	15	1.2								
12											

VALUE error: The VALUE error indicates a problem with the formula or cells that it references, often due to incompatible data types. Resolve that now by following these steps:

1. Locate the reference cell that contains the **#VALUE!** error message.
2. Select the cell with the **VALUE** error.

3. Check the formula in cell **G3** and identify where incompatible data types are being used. The formula bar shows the expression `=B3*C3`. When you check cells **B3** and **B4**, you'll notice that cell **B3** contains the word **text** instead of a value.
4. You'll need to adjust the formula or ensure that the referenced cells contain compatible data. In this case, enter a value into cell **B3**. In cell **B3**, enter **50** and press **Enter**.

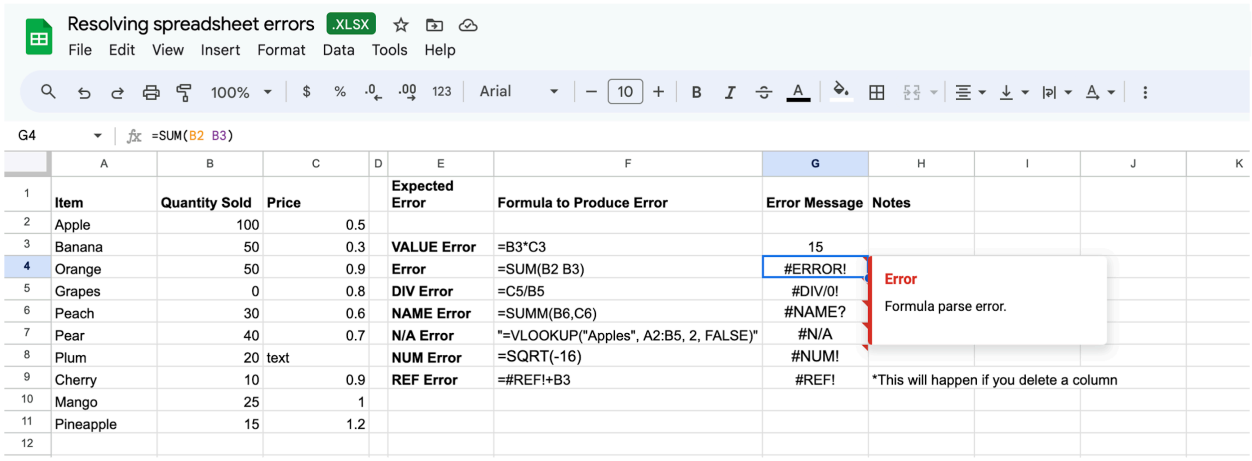


	A	B	C	D	E	F	G	H	I	J	K
1	**Item**	**Quantity Sold**	**Price**		**Expected Error**	**Formula to Produce Error**	**Error Message**	**Notes**			
2	Apple	100	0.5								
3	Banana	50	0.3		**VALUE Error**	=B3*C3	#VALUE!				
4	Orange	50	0.9		**Error**	=SUM(B2 B3)	#ERROR!				
5	Grapes	0	0.8		**DIV Error**	=C5/B5	#DIV/0!				
6	Peach	30	0.6		**NAME Error**	=SUMM(B6,C6)	#NAME?				
7	Pear	40	0.7		**N/A Error**	"=VLOOKUP("Apples", A2:B5, 2, FALSE)"	#N/A				
8	Plum	20	text		**NUM Error**	=SQRT(-16)	#NUM!				
9	Cherry	10	0.9		**REF Error**	=#REF!+B3	#REF!	*This will happen if you delete a column			
10	Mango	25	1								
11	Pineapple	15	1.2								
12											

5. Cell **G3** now shows the number 15 instead of the **VALUE** error, enabling the formula to run successfully.

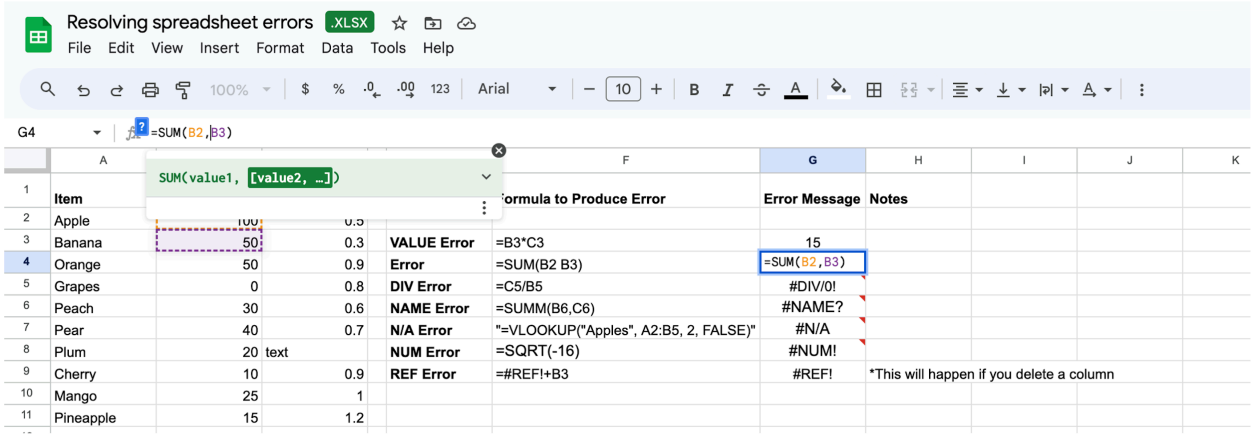
ERROR (parsing error): The parsing error occurs when a spreadsheet can't interpret the formula as it is input, often due to syntax issues. Remedy that now:

1. Locate the reference cell that contains the **#ERROR!** error message.
2. Select the cell with the **ERROR** error.
3. Select cell **G4** and check the formula for syntax errors. Common errors are missing operators or commas.
4. The expression in the formula bar shows that the `SUM` function being performed is missing a comma.



	A	B	C	D	E	F	G	H	I	J	K
1	**Item**	**Quantity Sold**	**Price**		**Expected Error**	**Formula to Produce Error**	**Error Message**	**Notes**			
2	Apple	100	0.5								
3	Banana	50	0.3		**VALUE Error**	=B3*C3	15				
4	Orange	50	0.9		**Error**	=SUM(B2 B3)	#ERROR!				
5	Grapes	0	0.8		**DIV Error**	=C5/B5	#DIV/0!				
6	Peach	30	0.6		**NAME Error**	=SUMM(B6,C6)	#NAME?				
7	Pear	40	0.7		**N/A Error**	"=VLOOKUP("Apples", A2:B5, 2, FALSE)"	#N/A				
8	Plum	20	text		**NUM Error**	=SQRT(-16)	#NUM!				
9	Cherry	10	0.9		**REF Error**	=#REF!+B3	#REF!	*This will happen if you delete a column			
10	Mango	25	1								
11	Pineapple	15	1.2								
12											

5. In the formula bar, delete the space between **B2** and **B3** and enter a comma (,), then press **Enter**.

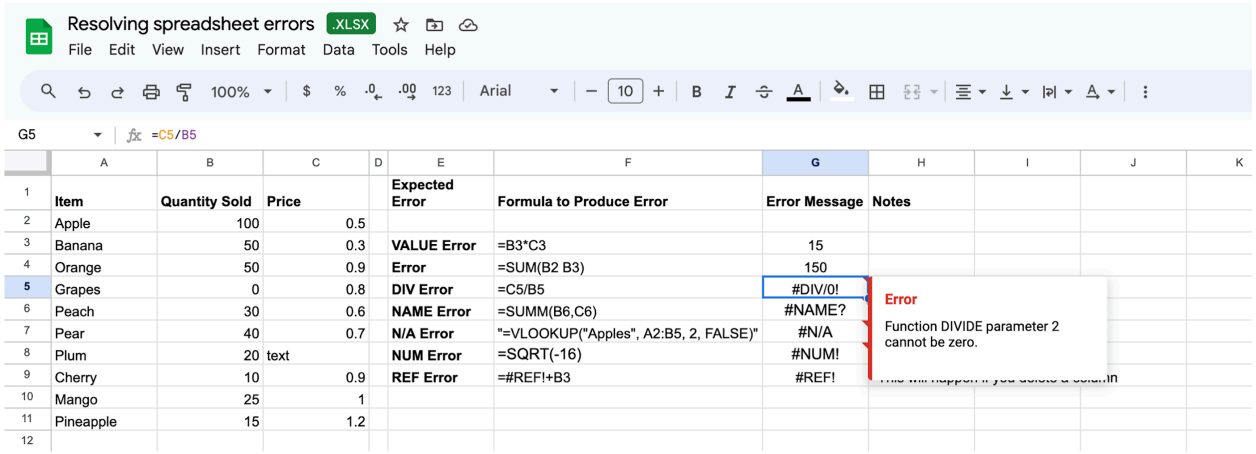


	A				F	G	H	I	J	K
1	Item				ormula to Produce Error	Error Message	Notes			
2	Apple	100	0.5							
3	Banana	50	0.3	VALUE Error	=B3*C3	15				
4	Orange	50	0.9	Error	=SUM(B2 B3)	=SUM(B2,B3)				
5	Grapes	0	0.8	DIV Error	=C5/B5	#DIV/0!				
6	Peach	30	0.6	NAME Error	=SUMM(B6,C6)	#NAME?				
7	Pear	40	0.7	N/A Error	"=VLOOKUP("Apples", A2:B5, 2, FALSE)"	#N/A				
8	Plum	20	text	NUM Error	=SQRT(-16)	#NUM!				
9	Cherry	10	0.9	REF Error	=#REF!+B3	#REF!	*This will happen if you delete a column			
10	Mango	25	1							
11	Pineapple	15	1.2							

6. Cell **G4** now shows the value 150 instead of the error message, fixing the problem.

DIV error: The **DIV** error occurs when you try to divide a number by zero or an empty cell. Solve this problem with these steps:

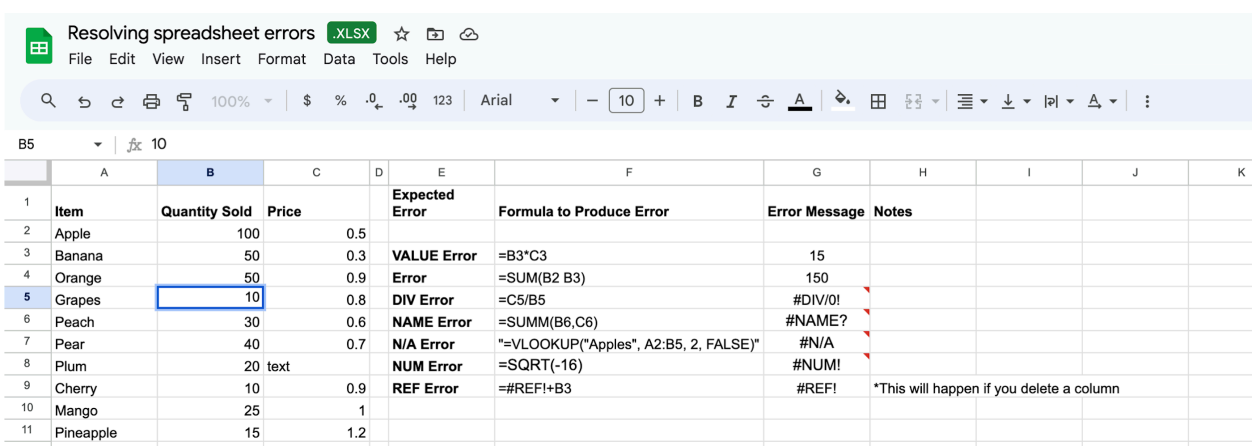
1. Locate the reference cell that contains the **#DIV/0!** error message.
2. Select the cell with the **DIV** error.
3. Check the formula in cell **G5** to identify where the division by zero is occurring.



	A	B	C	D	E	F	G	H	I	J	K
1	**Item**	**Quantity Sold**	**Price**		**Expected Error**	**Formula to Produce Error**	**Error Message**	**Notes**			
2	Apple	100	0.5								
3	Banana	50	0.3		**VALUE Error**	=B3*C3	15				
4	Orange	50	0.9		**Error**	=SUM(B2 B3)	150				
5	Grapes	0	0.8		**DIV Error**	=C5/B5	#DIV/0!				
6	Peach	30	0.6		**NAME Error**	=SUMM(B6,C6)	#NAME?				
7	Pear	40	0.7		**N/A Error**	"=VLOOKUP("Apples", A2:B5, 2, FALSE)"	#N/A				
8	Plum	20	text		**NUM Error**	=SQRT(-16)	#NUM!				
9	Cherry	10	0.9		**REF Error**	=#REF!+B3	#REF!				
10	Mango	25	1								
11	Pineapple	15	1.2								
12											

4. The expression in the formula bar shows that C5 is being divided by cell B5, meaning that B5 contains the value 0.

5. Modify the formula to avoid dividing by zero. Select cell **B5** and enter **10**, then press **Enter**.

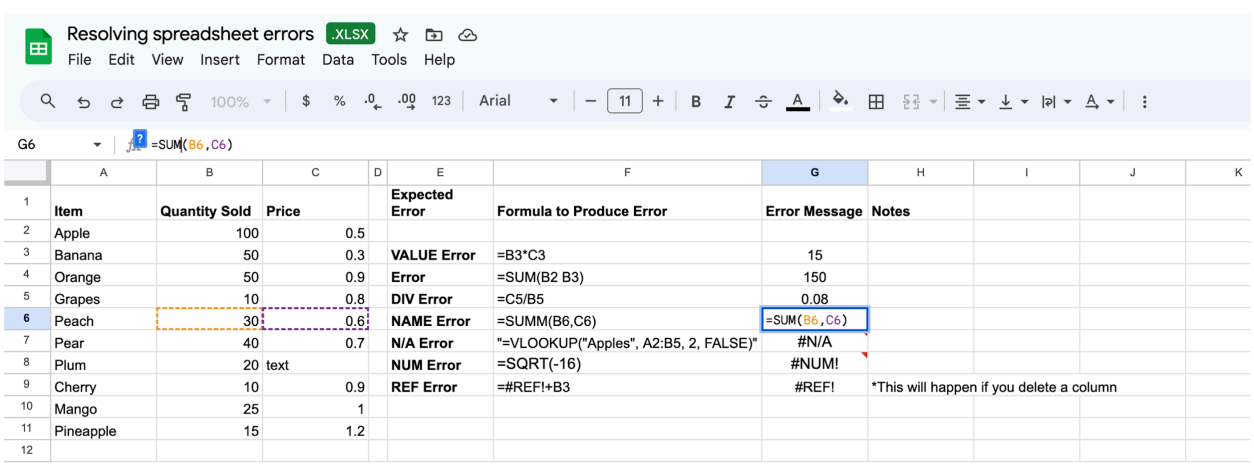


	A	B	C	D	E	F	G	H	I	J	K
1	**Item**	**Quantity Sold**	**Price**		**Expected Error**	**Formula to Produce Error**	**Error Message**	**Notes**			
2	Apple	100	0.5								
3	Banana	50	0.3		**VALUE Error**	=B3*C3	15				
4	Orange	50	0.9		**Error**	=SUM(B2 B3)	150				
5	Grapes	10	0.8		**DIV Error**	=C5/B5	#DIV/0!				
6	Peach	30	0.6		**NAME Error**	=SUMM(B6,C6)	#NAME?				
7	Pear	40	0.7		**N/A Error**	"=VLOOKUP("Apples", A2:B5, 2, FALSE)"	#N/A				
8	Plum	20	text		**NUM Error**	=SQRT(-16)	#NUM!				
9	Cherry	10	0.9		**REF Error**	=#REF!+B3	#REF!	*This will happen if you delete a column			
10	Mango	25	1								
11	Pineapple	15	1.2								

6. Cell G4 now shows the value **0.08** instead of the parsing error. Another error fixed!

NAME error: The **NAME** error occurs when the formula's name isn't recognized, usually due to a typo or incorrect function name.

1. Locate the reference cell that contains the **#NAME?** error message.
2. Select the cell with the **NAME** error.
3. Select cell **G6** and inspect the formula for any typographical errors or incorrect function names.
4. Delete the extra **M** in **SUMM** and press **Enter** to fix the typo in the formula bar. This is a toolbar that shows information contained in a cell.

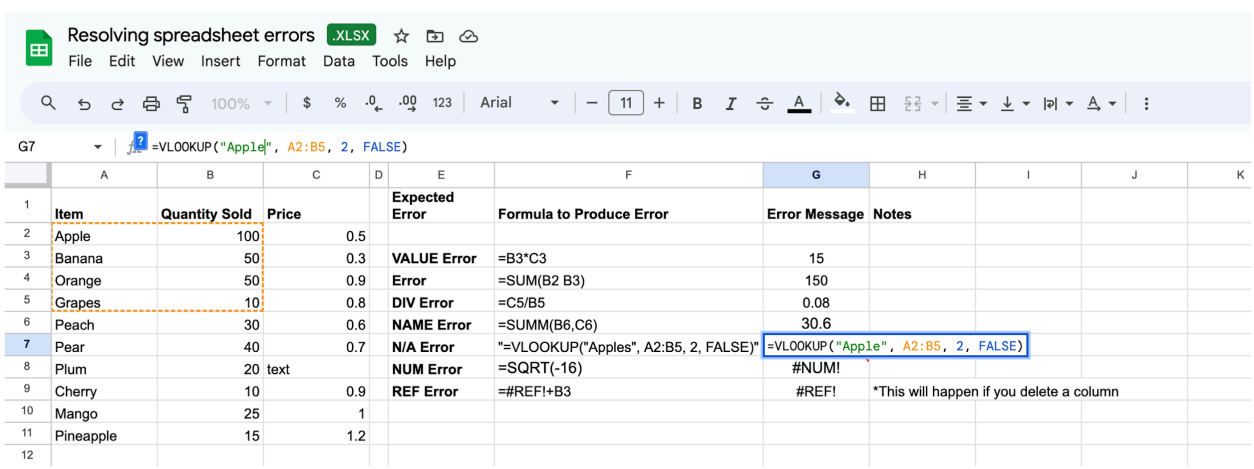


	Item	Quantity Sold	Price		Expected Error	Formula to Produce Error	Error Message	Notes
2	Apple	100	0.5					
3	Banana	50	0.3		**VALUE Error**	=B3*C3	15	
4	Orange	50	0.9		**Error**	=SUM(B2 B3)	150	
5	Grapes	10	0.8		**DIV Error**	=C5/B5	0.08	
6	Peach	30	0.6		**NAME Error**	=SUMM(B6,C6)	=SUM(B6,C6)	
7	Pear	40	0.7		**N/A Error**	"=VLOOKUP("Apples", A2:B5, 2, FALSE)"	#N/A	
8	Plum	20	text		**NUM Error**	=SQRT(-16)	#NUM!	
9	Cherry	10	0.9		**REF Error**	=#REF!+B3	#REF!	*This will happen if you delete a column
10	Mango	25	1					
11	Pineapple	15	1.2					

5. Cell **G6** now shows the value **30.6** instead of the **NAME** error, addressing the common spreadsheet error.

N/A error: The **N/A** error appears when the data referenced in your formula can't be found by the spreadsheet, often in lookup functions like `VLOOKUP`.

1. Locate the reference cell that contains the **#N/A** error message.
2. Select the cell with the **N/A** error.
3. Select cell **G7** and check the formula in that cell. Ensure the data being looked up exists in the specified range. The formula bar shows that "Apples" is being searched in the dataset, but "Apples" doesn't exist in column A. In the formula bar, delete the **s** so it matches the title **Apple** shown in cell **A2**. Press **Enter**.

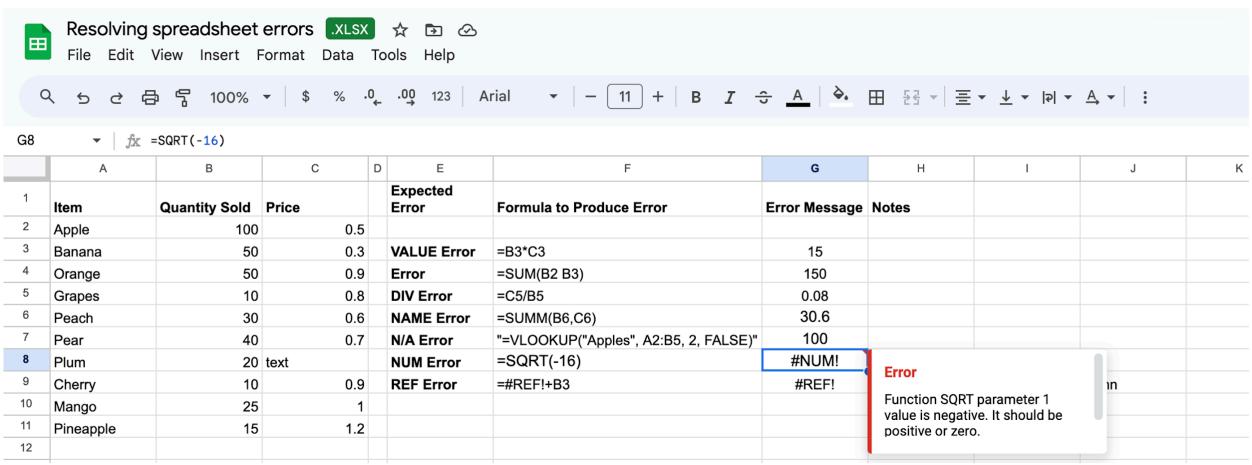


	A	B	C	D	E	F	G	H	I	J	K
1	Item	Quantity Sold	Price		Expected Error	Formula to Produce Error	Error Message	Notes			
2	Apple	100	0.5								
3	Banana	50	0.3		VALUE Error	=B3*C3	15				
4	Orange	50	0.9		Error	=SUM(B2 B3)	150				
5	Grapes	10	0.8		DIV Error	=C5/B5	0.08				
6	Peach	30	0.6		NAME Error	=SUMM(B6,C6)	30.6				
7	Pear	40	0.7		N/A Error	"=VLOOKUP("Apples", A2:B5, 2, FALSE)"	=VLOOKUP("Apple", A2:B5, 2, FALSE)				
8	Plum	20	text		NUM Error	=SQRT(-16)	#NUM!				
9	Cherry	10	0.9		REF Error	=#REF!+B3	#REF!	*This will happen if you delete a column			
10	Mango	25	1								
11	Pineapple	15	1.2								
12											

4. Cell G7 now shows the value 100 instead of the **N/A** error. Great work!

NUM error: The **NUM** error indicates that the formula's calculation can't be performed as specified, often because the data doesn't make sense for that calculation.

1. Locate the reference cell that contains the **#NUM** error message.
2. Select the cell with the **NUM** error.
3. Check the formula and identify why the calculation is not feasible.



	A	B	C	D	E	F	G	H	I	J	K
1	**Item**	**Quantity Sold**	**Price**		**Expected Error**	**Formula to Produce Error**	**Error Message**	**Notes**			
2	Apple	100	0.5								
3	Banana	50	0.3		**VALUE Error**	=B3*C3	15				
4	Orange	50	0.9		**Error**	=SUM(B2 B3)	150				
5	Grapes	10	0.8		**DIV Error**	=C5/B5	0.08				
6	Peach	30	0.6		**NAME Error**	=SUMM(B6,C6)	30.6				
7	Pear	40	0.7		**N/A Error**	"=VLOOKUP("Apples", A2:B5, 2, FALSE)"	100				
8	Plum	20	text		**NUM Error**	=SQRT(-16)	#NUM!				
9	Cherry	10	0.9		**REF Error**	=#REF!+B3	#REF!				
10	Mango	25	1								
11	Pineapple	15	1.2								
12											

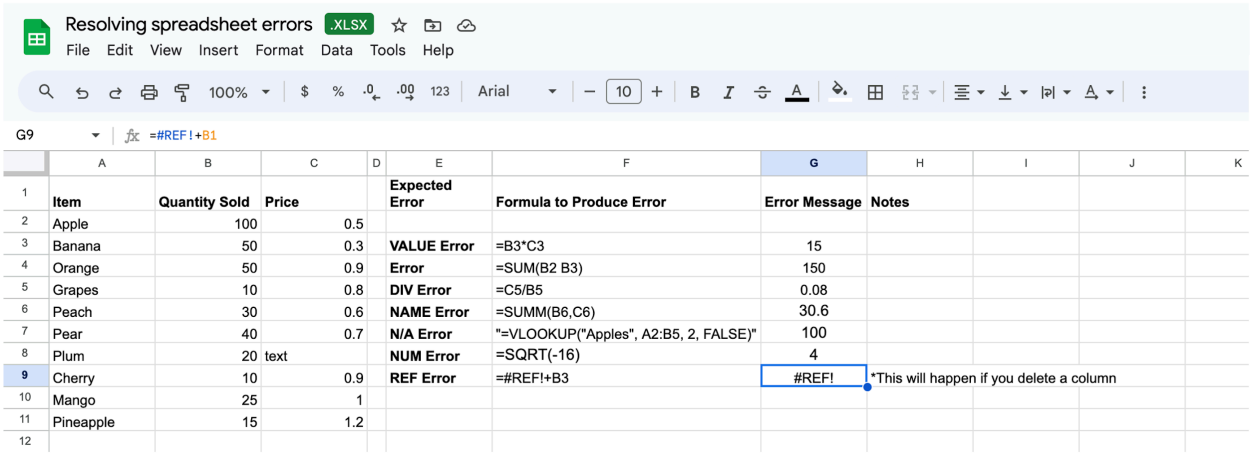
4. The spreadsheet can't take the square root of a negative number. (See note.) So, in the formula bar, delete the **-** (minus) and press **Enter**.

5. Cell **G8** now shows the value **4** instead of the **NUM** error, solving yet another error!

Note: In mathematics, it's possible to take the square root of a negative number, as the result is not a real number; it’s considered imaginary. That's beyond the scope of this course, but just know that spreadsheet applications will provide **NUM** errors when certain mathematical conditions are met.

REF error: The **REF** error occurs when cells being referenced in a formula have been deleted, making the formula unable to perform the calculation.

1. Locate the reference cell that contains the **#REF!** error message.
2. Select the cell with the **REF** error.
3. Select cell **G9** and check the formula to find out which reference cells have been deleted or modified.



	A	B	C	D	E	F	G	H	I	J	K
1	**Item**	**Quantity Sold**	**Price**		**Expected Error**	**Formula to Produce Error**	**Error Message**	**Notes**			
2	Apple	100	0.5								
3	Banana	50	0.3		**VALUE Error**	=B3*C3	15				
4	Orange	50	0.9		**Error**	=SUM(B2 B3)	150				
5	Grapes	10	0.8		**DIV Error**	=C5/B5	0.08				
6	Peach	30	0.6		**NAME Error**	=SUMM(B6,C6)	30.6				
7	Pear	40	0.7		**N/A Error**	"=VLOOKUP("Apples", A2:B5, 2, FALSE)"	100				
8	Plum	20	text		**NUM Error**	=SQRT(-16)	4				
9	Cherry	10	0.9		**REF Error**	=#REF!+B3	#REF!	*This will happen if you delete a column			
10	Mango	25	1								
11	Pineapple	15	1.2								
12											

4. In this case, update the reference cells by deleting the expression in the formula bar and typing `=B2+B3`, then press **Enter**.

5. The value in cell **G9** now shows **150**—no more **REF** error!

Note: When you have a cell or column referenced, then delete the column, sometimes you'll receive the **#REF!** error. Making sure the cells in your functions are correct is crucial for a data analyst, as modifications to spreadsheets can make your calculations produce errors or become incorrect.

Pro Tip: Save the activity template

Be sure to save a copy of the spreadsheet template you used to complete this activity. You can use it for further practice or to help you work through your thought processes for similar tasks in a future data analyst role.

1.

Question 1

Reflection

In this exercise, you resolved common spreadsheet errors. Now, take a few minutes to reflect on this activity.

- What was the most challenging error you encountered, and how did you successfully resolve it?
- How does error resolution contribute to data accuracy? Why is this so important to data analysis and the reliability of results?
- How do you plan to apply the error identification and resolution skills you've gained in this activity to your future data analysis projects?

In the space provided below, write 2-3 sentences (40-60 words) to respond to each of these questions and your experience resolving errors.

Answer:
For my turn, the most challenging error is the REF error.
Ref error by deleting row or column, So it is difficult to understand which row or column is deleted; after finding the proper row or column that was deleted, then I resolved this problem.

If there was an error in the data, then we won't get the proper results, so here comes data accuracy. So to get the proper result, we have to focus on data accuracy; to do so, we have to be compulsory on error resolution.
Inappropriate or missing data doesn't give the appropriate results, and if the results are not proper, then insights will be valueless and the data-driven decision will also be valueless. So in the data analysis, error-free data will give proper results and can give data-driven decisions, which is the most important part of data analysis.

If we work with spreadsheets, there are mainly six types of errors that have occurred, i.e., VALUE Error, Error, DIV Error, NAME Error, N/A Error, NUM Error, and REF Error. Each type of error occurs with a particular type of fault, which we learned from the course; now we can identify it in the spreadsheet and remove it if any error occurs. In the future project we will apply this knowledge when we notice this type of error will occur.

Correct

Great job diving deep into spreadsheet error resolution challenges and nuances! A well-rounded reflection would include specific errors you encountered and how they emphasized the importance of accurate data analysis.

Proactively addressing spreadsheet errors equips you with invaluable data analytics skills. After some practice and time, these skills will become second nature, enabling you to enhance data accuracy even further. As you move forward, remember to apply this knowledge to your future projects, ensuring precise and reliable data analysis.